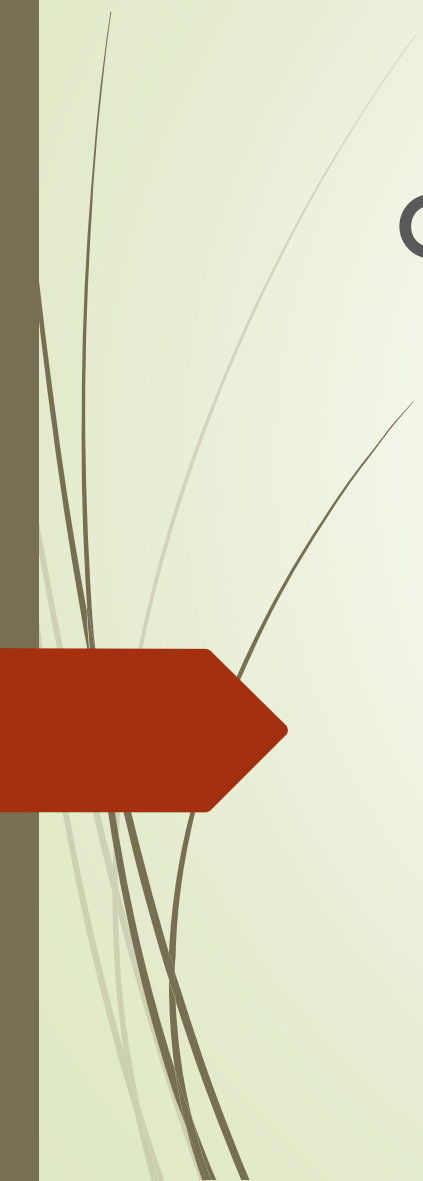




Global Position Systems Operations - Fundamentals of GPS, Error Sources, and Differential Corrections



Objectives

Objectives:

- Understand various errors and where they come from
- Understand differential corrections: various correction methods

GPS

GPS receiver's errors vary depending on

- Number of satellites in view and satellite geometry
 - Where, when, weather, etc.
- Quality of the range and range rate measurements
 - Errors in navigation message parameters
 - Signal transmission time
 - Propagation delay in the ionosphere and the troposphere
 - Signal distortion
 - Receiver noise
 - Etc.

GPS

GPS positioning accuracy

- Horizontal 100m and vertical 156m: 1990s, SA (selective availability) is the major source of error
- Horizontal 10m and vertical 15m: after 2001 – post SA

DGPS

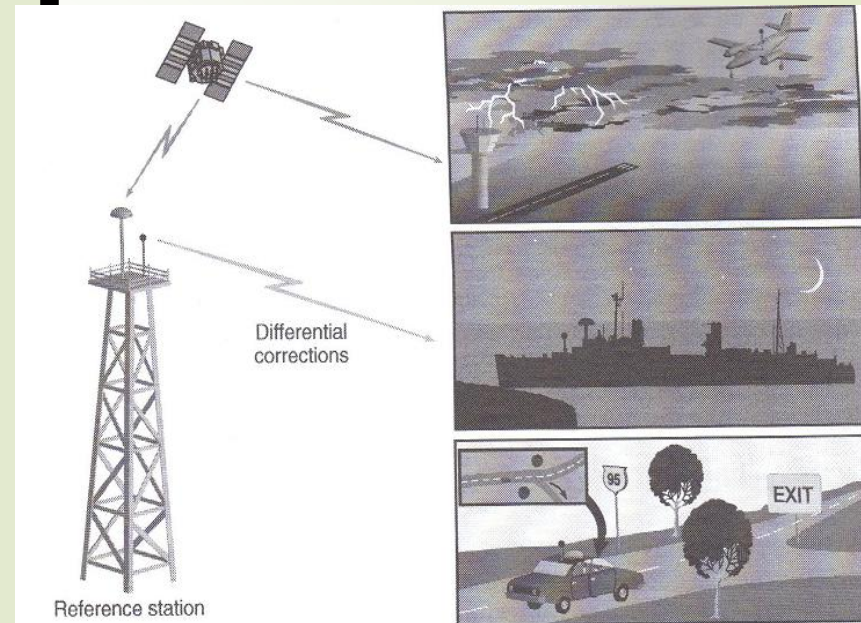
Differential GPS (DGPS)

- GPS's estimates can be improved by reducing measurement errors and/ or improving satellite geometry.
- We can estimate the error in a measurement if the receiver location or nearby station is known. This kind error estimate is referred to as differential corrections. That is Differential GPS (DGPS).

Local-area Differential GPS (LADGPS)

- A known reference station is set up to send error estimate to GPS users around the area and within this area all the users can use this error to do differential corrections.
- Accuracy: can be meter-level, depending on how close the user to the reference station.
- One reference station usually has signal ranges of about 400 km.
- 50 stations for the U.S. Coast Guard. Accuracy 1-3 meters
- Nationwide Differential GPS (NDGPS)

DGPS



Misra and Enge,
2011

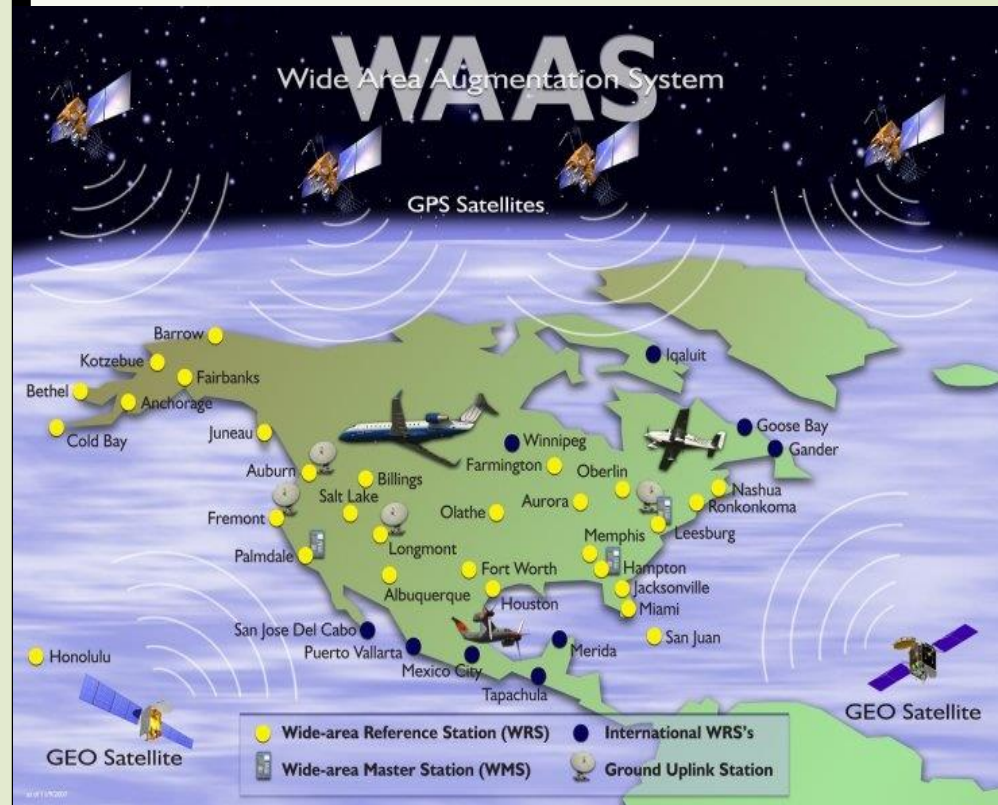
WADGPS

Wide-area Differential GPS

(WADGPS) – a centralized system to provide differential corrections via communication satellites.

(Federal Aviation Admi.)

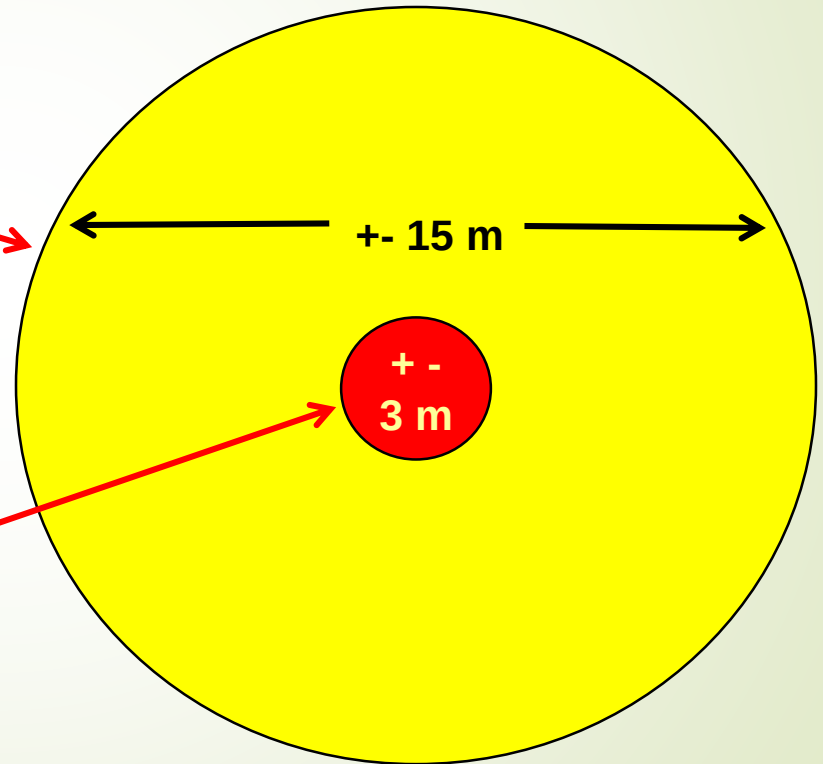
FAA's Wide Area Augmentation System (WAAS) for North America: a real-time differential correction service and is based on a network of approximately 25 ground reference stations that cover a very large service area.



Wide Area Augmentation System (WAAS)

With Selective Availability set to zero, and under ideal conditions, a GPS receiver without WAAS can achieve 15-meter accuracy most of the time.*

Under ideal conditions, a WAAS equipped GPS receiver can achieve 3-meter accuracy 95% of the time.*



* Precision depends on good satellite geometry, open sky view, and no user-induced errors.



DGPS positioning accuracy

Positioning accuracy varies depending on with or without differential corrections, code phase and carrier phase measurements, LADGPS and WADGPS, GPS types (survey, mobile mapping, and portable GPS), SA on and off.

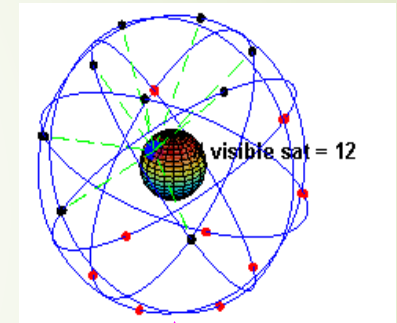
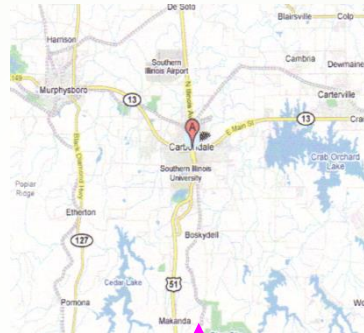
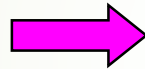
Errors: centimeter-level to 100m



How GPS turn-by-turn navigation works

OnStar: turn-by-turn navigation:

How does it work? Combination of GIS with remote sensing and GPS for navigation:



Navigation systems

- GPS
 - USA
 - Global Positioning System
 - First launch 1978 and Operational 1993
- GLONASS
 - Russia
 - Global Navigation Satellite System
 - 1988
- Galileo
 - European Union
 - 2005
- BeiDuo
 - China
 - 2000

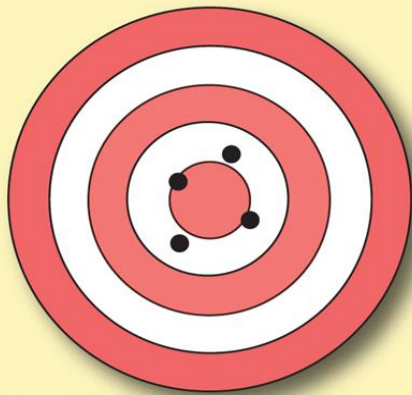
Error, bias, accuracy, precision

Accuracy: means correctness and agreement between an estimate and reference (that is, correct) value.

Bias: means differences between estimates and observations (reference values), tending to persist over a period of time.

Measurement errors: means noise or bias, averaging out to zero

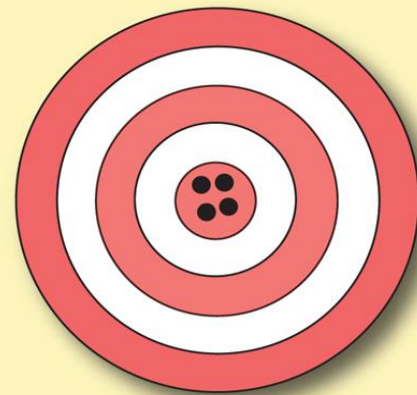
Accuracy versus Precision



a. High Accuracy,
Low Precision



b. Low Accuracy,
High Precision

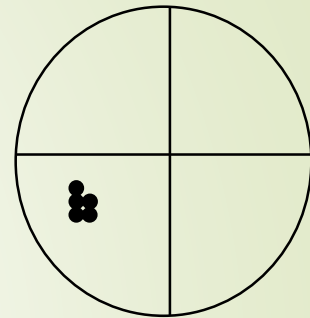


c. High Accuracy,
High Precision

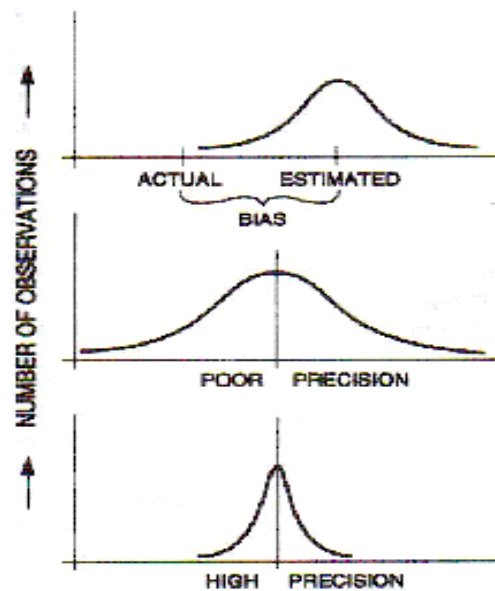
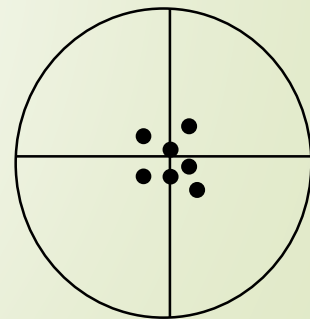
Error, bias, accuracy, precision

Precision: If using an instrument to measure a tree height, we repeat measuring it several times, the height values are close to each other in **(A)**, this instrument is precise, but not accurate. On the other hand, the height values are spatially distributed as in **(B)**, but not close to each other, this instrument is not precise, but the results are accurate.

(A)



(B)



Error sources and models

Error groups

- Errors in the parameters broadcast by the satellite – control responsible
- Errors associated with the propagation medium that affect the signal traveling time
- Noise of the receiver: interference from signals reflected from surfaces nearby

Error sources and models

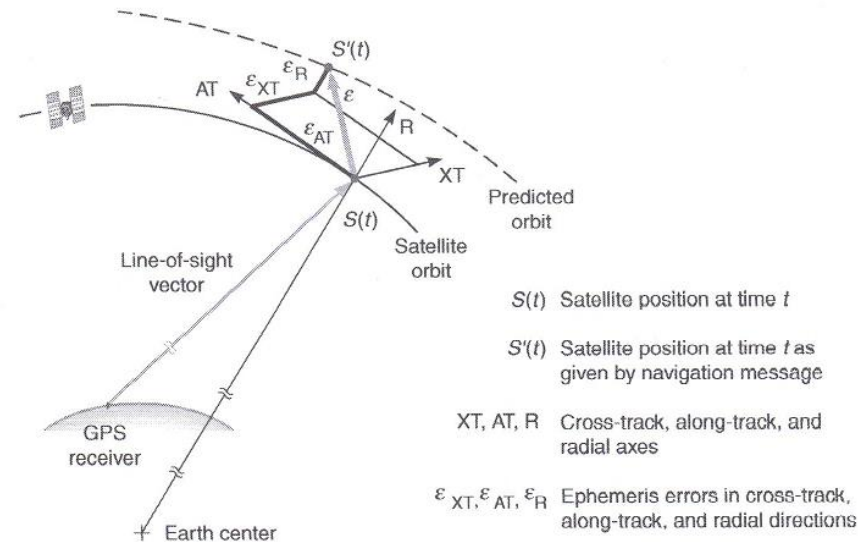
Error groups

- Control segment errors
- Signal propagation modeling error
 - Signal refraction, wave propagation, and dispersive media
 - Ionosphere delay
 - Troposphere delay
- Measurement errors
 - Receiver noise
 - Multipath
 - Empirical data
- User range error
- Selective availability
 - But, turned off in 2000

Control segment errors

- Satellite clock and ephemeris parameters broadcast by the satellites are derived by the GPS monitor stations.
- Previously, the ranging error due to Satellite clock and ephemeris parameters is limited to 6m for PPS, and now limited to 3m.
- The ephemeris error is divided into three components relative to the satellite orbits:
 - Radial, along-track and cross-track
- The error in the pseudorange measurement is mostly depends on radial component.

Error sources and models



Misra and Enge, 2011

Error sources and models

Signal propagation modeling errors

- GPS signals are affected by the propagation medium through which the signals travel from satellites to the receivers.
- Refraction: Atmosphere changes the speed and direction of signals, thus changes the apparent time and the pseudorange measurements.
- Ionospheric delay: 50km to 1000km, ionized gases, changing the speed of the signals, the delay can be estimated using Dual-Frequency measurements.
- **Tropospheric delay:** less than 20km, dry gases (N₂ and O₂) and water vapor, lowering the speed of the GPS signals, the delay can be estimated using tropospheric models.



Error sources and models

Measurement errors

- **Receiver noise:** random measurement noise - smudges the signal, affecting precision of the code and carrier phase measurement.
 - Mainly introduced by antenna, amplifier, cables, the receiver, multi-access noise (i.e., interference from other GPS units), and signal quantization noise.
- **Multipath:** signals from two or more paths - introducing interfering signals and changing the phase.
 - Structures such as buildings in the vicinity

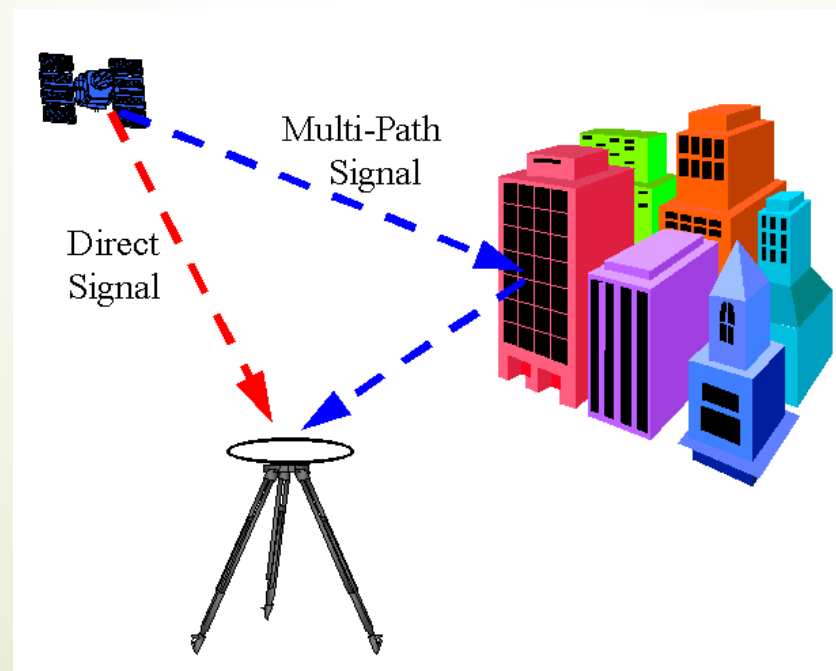
Error sources and models

Multipath



What is multi-path?

- Multi-path occurs when signals are reflected off of objects such as trees or buildings.
- These reflections delay the signal before it reaches the antenna and throw off the range calculations.
- Multi-path is the greatest source of error in forestry settings and the most difficult to combat.



Error sources and models

User range error (URE) <http://www.gpshacks.com/GPS-Error>

Error source	Potential error	Typical error
Ionosphere	5.0 meters	0.4 meters
Troposphere	0.5 meters	0.2 meters
Ephemeris data	2.5 meters	0 meters
Satellite clock drift	1.5 meters	0 meters
Multipath	0.6 meters	0.6 meters
Measurement noise	0.3 meters	0.3 meters
Total	~15 meters	~10 meters

Error corrections: differential GPS (DGPS)

Idea behind DGPS

- The errors associated with satellite clock, ephemeris (satellite position), atmospheric propagation medium, are similar for all the users within an area of hundreds of kilometers.
- The errors are spatially and temporally autocorrelated. The closer the users, the more similar the errors.
- The combined effect of the errors can be estimated for each satellite and used to improve the quality of GPS estimates for all the users within this area.
- **On the other hand, the errors due to the receiver noise and multipath can not be corrected by DGPS!**

Error corrections: differential GPS (DGPS)

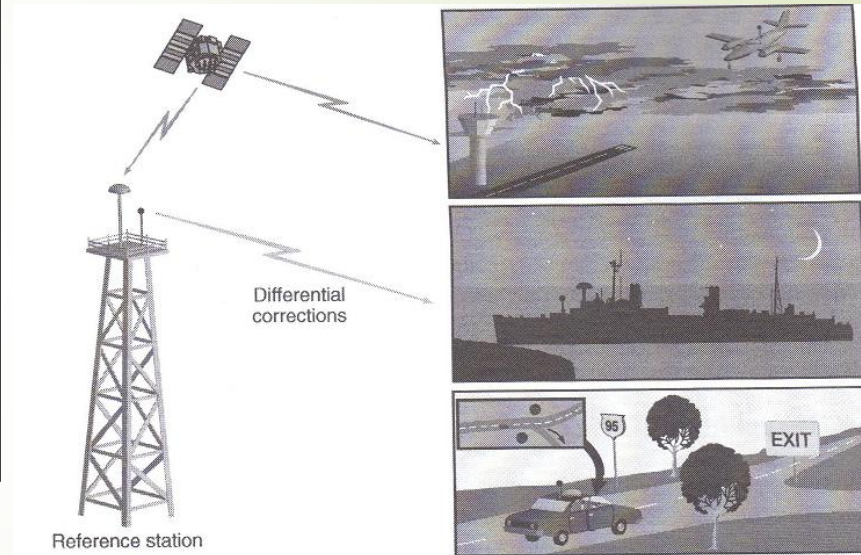
Differential GPS (DGPS):

- We can estimate the error in a measurement if a reference receiver or station is known.
- This error can be then used for the users within the area covered by the reference receiver or station.

Error corrections: differential GPS (DGPS)

Two questions associated with DGPS:

1. What is the appropriate size of the area a reference station can serve?
2. How often the differential corrections can be provided?



Error corrections: differential GPS (DGPS)

- Radio Technical Commission for Maritime Services (RTCM) is widely used to transmit the differential corrections.
- US Coast Guard's Maritime DGPS: broadcast differential corrections in the RTCM SC-104 at a rate of 100-200bps. Operated in 1999
- 50 stations and accuracy 1-3 meters
- Nationwide DGPS was developed by adding reference stations, leading to each station covering a signal range of 400 km. 48 states are covered:
 - Operated in 2004 and accuracy 1-3m.

Error corrections: differential GPS (DGPS)

Wide-Area DGPS (WADGPS):

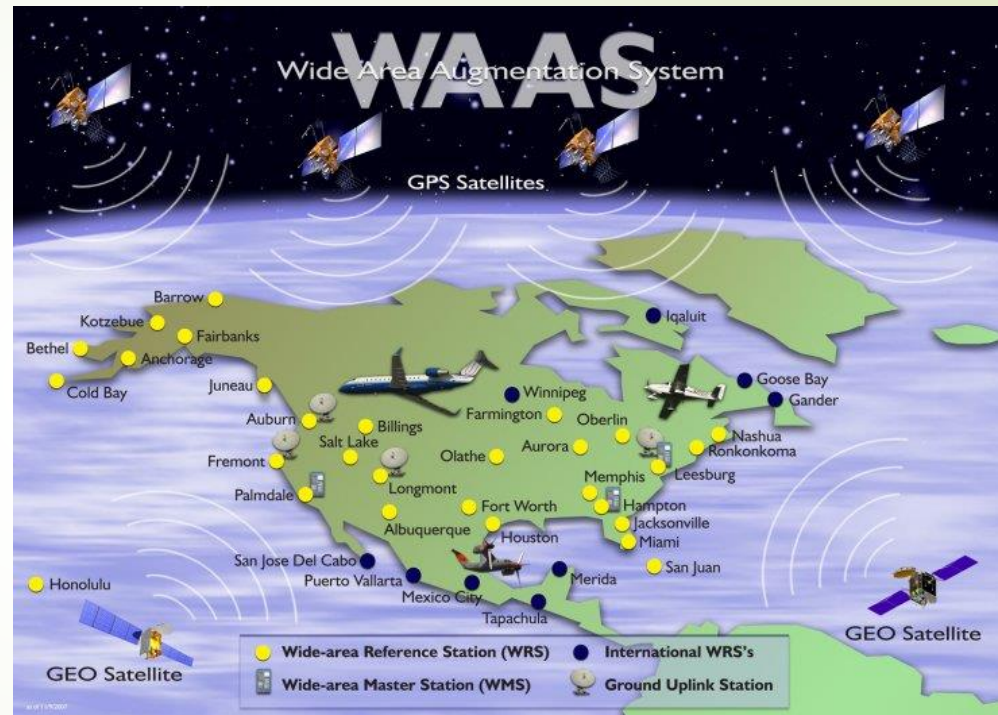
- A large number of reference stations are deployed over a large region.
- Geostationary satellites or a network of FM stations are used to broadcast the differential corrections.
- Instead of lumped error by LADGPS, WADGPS broadcasts differential correction vectors (that is, an error for each of the sources).

Error corrections: differential GPS (DGPS)

Wide-Area DGPS

(WADGPS):

- Federal Aviation Admi. (FAA)'s Wide Area Augmentation System (WAAS) for North America: a real-time differential correction service and 25 ground reference stations.
- The differential corrections (clock, ephemeris, iono.) are coded as a 250 bps. A GPS receiver would require software modification in order to get these data.



<http://www.nstb.tc.faa.gov/>

Error corrections: differential GPS (DGPS)

Wide-Area DGPS (WADGPS):

- General name: Satellite Based Augmentation System (SBAS)
- European Geostationary Navigation Overlay System (EGNOS) in 2009
- Multifunction Transportation Satellite (MTSAT) based Satellite Augmentation System in Japan in 2007.

Error corrections: differential GPS (DGPS)

Wide-Area DGPS (WADGPS):

- NASA's Global Differential GPS (GDGPS)
 - Providing decimeter-level positioning accuracy
 - Provide accurate, real-time estimates of GPS ephemeris and clock parameters.
 - Transmit the corrections over the internet:
- <http://www.gdgps.net>



GPS applications in reclamation

- Locating points, roads, etc.
- Verify stream buffer zones, minesite boundaries, etc.
- Mapping and calculating minesite areas
- Etc.